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Sub ject : DSA

Unit 3, S2, SLO2 - Stack Implementation Using Linked Lists

Push Operation - Algorithm to insert an element VAL in a linked stack
Step 1: Allocate memory for the new node and name it as NEW_NODE
Step 2: SET NEW_NODE -> DATA = VAL
Step 3: IF TOP = NULL
SET NEW_NODE -> NEXT = NULL
SET TOP = NEW_NODE
ELSE
SET NEW_NODE -> NEXT = _________
SET TOP = ____________
[END OF IF]
Step 4: END

Answer: Step 3: SET NEW_NODE -> NEXT = TOP
SET TOP = NEW_NODE
Step 4: END

Pop Operation - Algorithm to delete an element from a linked stack
Step 1: IF TOP = NULL
PRINT UNDERFLOW
Goto Step 5
[END OF IF]
Step 2: SET PTR = TOP
Step 3: SET TOP = TOP -> _________
Step 4: FREE _________
Step 5: END

Answer: Step 3: SET TOP = TOP -> NEXT
Step 4: FREE PTR
Step 5: END

Peek Operation - Algorithm for peek in linked stack
Step 1: IF TOP = NULL
PRINT STACK IS EMPTY
Goto Step 3
[END OF IF]
Step 2: RETURN TOP -> _________
Step 3: END

Answer: Step 2: RETURN TOP -> DATA
Step 3: END